

SANZIDA MOJIB LUNA

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I am a **Human-Computer Interaction (HCI)** researcher specializing in inclusive design and adaptive technologies, particularly within extended reality (XR). My research investigates the lived experiences of underrepresented communities, including Deaf and Hard of Hearing (DHH) users and older adults, through methods such as participatory design, user studies, and longitudinal engagement, with the aim of informing the design of accessible and ethically grounded digital technologies.

EDUCATION

Rochester Institute of Technology (RIT)

PhD in Computing and Information Sciences

Advisor: Dr. Konstantinos Papangelis

Aug 2022 - Present

Military Institute of Science and Technology (MIST)

BSc in Computer Science and Engineering

Feb 2016 - Dec 2019

RESEARCH EXPERIENCE

Rochester Institute of Technology (RIT)

Graduate Research Assistant

Niantic X RIT Geo Games and Media Research Lab

Aug 2022 - Present

- Led and contributed to multiple XR accessibility projects focused on Deaf and Hard of Hearing (DHH) users, including co-located multiplayer AR games and instructional AR environments
- Designed and conducted user studies with 25+ DHH and hearing participants using mixed methods, including contextual inquiries, think-aloud protocols, and semi-structured interviews
- Developed interview protocols, survey instruments, and recruitment materials in alignment with ethical research practices
- Coordinated accessible study environments by working closely with ASL interpreters and implementing inclusive communication strategies
- Employed qualitative analysis techniques to identify experiential themes and behavioral patterns in immersive XR environments
- Conducted a systematic review of 553 papers and developed an analytical framework to examine accessibility trends, demographic representation, and inclusive design strategies in AR research
- Conducted expert interviews and synthesized findings with literature to develop structured guidelines for ethical and inclusive XR research with DHH participants
- Served as first author on multiple academic publications based on these projects

Augment Therapy

Research Collaborator

Aug 2024 - Present

- Led mixed-methods research investigating user motivation, engagement, and accessibility in mobile mixed reality (MR) health-focused applications
- Designed and conducted a 10-day deployment study of an MR-based therapeutic game, collecting behavioral metrics, daily surveys, and conducting in-depth interviews
- Applied thematic analysis to identify barriers to engagement and co-organized participatory design workshops to inform accessibility-centered interaction strategies
- Designed and evaluated a prototype based on study findings; collaborated with product design teams to translate research insights into actionable design refinements
- Conducted longitudinal studies with older adults (65+) to explore the impact of social exergaming in MR on motivation, well-being, and mental health
- Developed study protocol using a mixed-methods approach to track behavioral and emotional shifts over time

LIST OF PUBLICATIONS

1. **Sanzida Mojib Luna**, Jiangnan Xu, Elise Baron, Garreth W. Tigwell, and Konstantinos Papangelis. 2025. Motivation and Re-engagement in Mixed Reality: How Deaf and Hard of Hearing Users Experience a Mixed Reality Exergame. In The 27th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '25), October 26–29, 2025, Denver, CO, USA. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3663547.3759687>
— Accepted as Poster in ASSETS '25

2. Jiangnan Xu, **Sanzida Mojib Luna**, Garreth W. Tigwell, Nicolas Lalone, Michael Saker, Samuli Laato, John Dunham, Yihong Wang, Alan Chamberlain, and Konstantinos Papangelis. 2025. Understanding the Interplay Between the Digital and the Physical in Shared Augmented Reality Gaming: Probing through Urban Legends. *ACM Trans. Comput.-Hum. Interact.* Just Accepted (August 2025). <https://doi.org/10.1145/3749841>
— Accepted as Journal Article in *ACM Transactions on Computer-Human Interaction*
3. **Sanzida Mojib Luna**, Jiangnan Xu, Garreth W. Tigwell, Nicolas LaLone, V Saker, Alan Chamberlain, David I Schwartz, and Konstantinos Papangelis. 2025. Exploring Deaf And Hard of Hearing Peoples' Perspectives On Tasks In Augmented Reality: Interacting With 3D Objects And Instructional Comprehension. In *CHI Conference on Human Factors in Computing Systems (CHI '25)*, April 26-May 1, 2025, Yokohama, Japan. ACM, New York, NY, USA, 14 pages. <https://doi.org/10.1145/3706598.3713678>
— Accepted as Full Paper in *CHI '25*
4. **Sanzida Mojib Luna**, Garreth W. Tigwell, Konstantinos Papangelis, and Jiangnan Xu. 2024. Exploring Visual Scanning in Augmented Reality: Perspectives From Deaf and Hard of Hearing Users. In *The 26th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '24)*, October 27–30, 2024, St. John's, NL, Canada. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3663548.3688535>
— Accepted as Poster in *ASSETS '24*
5. **Sanzida Mojib Luna**, Jiangnan Xu, Konstantinos Papangelis, Garreth W. Tigwell, Nicolas Lalone, Michael Saker, Alan Chamberlain, Samuli Laato, John Dunham, and Yihong Wang. 2024. Communication, Collaboration, and Coordination in a Co-located Shared Augmented Reality Game: Perspectives From Deaf and Hard of Hearing People. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI '24)*. Association for Computing Machinery, New York, NY, USA, Article 36, 1–14. <https://doi.org/10.1145/3613904.3642953>
— Accepted as Full Paper in *CHI '24*
6. Jiangnan Xu, Konstantinos Papangelis, Garreth W. Tigwell, Nicolas Lalone, Pengyuan Zhou, Michael Saker, Alan Chamberlain, John Dunham, **Sanzida Mojib Luna**, and David Schwartz. 2024. Spatial Computing: Defining the Vision for the Future. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*. Association for Computing Machinery, New York, NY, USA, Article 583, 1–4. <https://doi.org/10.1145/3613905.3643978>
— Accepted as Extended Abstract in *CHI '24*
7. **Sanzida Mojib Luna**, Garreth W. Tigwell, Konstantinos Papangelis, and Jiangnan Xu. 2023. Communication and Collaboration Among DHH People in a Co-located Collaborative Multiplayer AR Environment. In *Proceedings of the 25th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '23)*. Association for Computing Machinery, New York, NY, USA, Article 64, 1–5. <https://doi.org/10.1145/3597638.3614479>
— Accepted as Poster in *ASSETS '23*
8. **Sanzida Mojib Luna**. 2023. DHH People in Co-located Collaborative Multiplayer AR Environments. In *Companion Proceedings of the Annual Symposium on Computer-Human Interaction in Play (CHI PLAY Companion '23)*. Association for Computing Machinery, New York, NY, USA, 344–347. <https://doi.org/10.1145/3573382.3616039>
— Accepted as Doctoral Consortium in *CHI PLAY '23*
9. Barsha, Farhat Lamia, Zarin Tasneem, **Sanzida Mojib**, Masuda Afrin, Nusrat Jahan, Marzouka Tasnim, Umma Habiba, and Muhammad Nazrul Islam. "An IoT-based automated door accessing system for visually impaired people." In *2019 IEEE International WIE Conference on Electrical and Computer Engineering (WIECON-ECE)*, pp. 1-4. IEEE, 2019. <https://doi.org/10.1109/WIECON-ECE48653.2019.9019945>
— Accepted as a conference paper

TEACHING EXPERIENCE

Course: GAME DESIGN & DEVELOPMENT II (IGME- 320)

Jan 2025 - May 2025

Teaching Assistant

Course Instructor: Dr. Konstantinos Papangelis

- Delivered a guest lecture on *Accessible Game Designs* and *User Research Methods*
- Mentored student teams in refining game concepts through iterative feedback and critical review
- Guided students in designing and developing user studies to evaluate their projects, emphasizing experimental design, ethical research practices, and inclusive evaluation strategies

PROFESSIONAL SERVICES

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| 1. Reviewer for <i>ACM CHI</i> Conference (full-length conference paper) | 2023-2025 |
| 2. Reviewer for <i>ACM CHI PLAY</i> , <i>AutomotiveUI</i> , <i>IMWUT</i> , <i>MobileHCI</i> , <i>INTERACT</i> , <i>TEI</i> Conference | 2023-2025 |
| 3. Guest Editor of a Special Issue on Spatial Computing for the journal <i>Behaviour & Information Technology</i> | 2024-2025 |
| 4. Led a Special Interest Group (SIG) at CHI'24 | 2024 |

5. Contributed to grant writing efforts for <i>Niantic X RIT Geo Games and Media Research Lab</i>	2023
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TRAVEL GRANTS

27th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS 2025) Presented poster (List of Publications #1) at ASSETS'25 in Denver, Colorado, USA.	2025
2024 Conference on Human Factors in Computing Systems (CHI 2024) Presented a full paper (List of Publications #6) and led a Special Interest Group (SIG) at CHI 2024 as a first-time in-person attendee in Honolulu, Hawai'i, USA.	2024
25th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS 2023) Presented poster (List of Publications #7) at ASSETS'23 as a first-time attendee in NYC, New York, USA.	2023

INVITED TALKS AND PRESENTATIONS

1. Guest lecturer at <i>IGME-704</i> at RIT	2025
2. Presented poster at ASSETS'25	2025
3. Presented full paper as a virtual presenter at CHI'25	2025
4. Presented research work at the Graduate Showcase at RIT	2025
5. Guest speaker at the University of California, Irvine Center for Hearing Research Journal Club	2025
6. Guest speaker at the Versatelist Podcast by Immersive Learning Research Network (ILRN)	2025
7. Guest Speaker at the <i>Research Potential Assessment</i> Q/A Session	2025
8. Presented poster at <i>a2ru 2024</i> at RIT	2024
9. Presented poster as a virtual presenter at ASSETS'24	2024
10. Presented full paper at CHI'24	2024
11. Presented poster at ASSETS'23	2023

AWARD AND HONORS

Nominated as one of the four applicants for Google PhD Fellowship by GCCIS, RIT	2025
Featured in RIT Magazine Spring 25 Issue for contributions to accessibility research in MAR	2025
Honorable Mention in 3MT Competition at RIT	2024
Special Recognition for Outstanding Review for <i>CHI PLAY Work In Progress</i>	2024
Best Paper at the WEICON-ECE 2019	2019

RELEVANT COURSEWORK

- Research Foundations
- Research Methods
- Future Interactions

SKILLS AND OTHER INFORMATION

Technical Skills: Java, Python, C/C++, HTML/CSS/JavaScript, MATLAB, MySQL, Oracle, Firebase.

Software Tools/Framework: Overleaf, MAXQDA, NVivo, Git, Weka, Eclipse, NetBeans, VS Code, Android Studio, Blender, Adobe Illustrator, Adobe Photoshop, Canva.

Hardware: Raspberry Pi, Arduino, PIC16F877A.

Language: English (fluent), Bengali (native), Hindi (conversational).